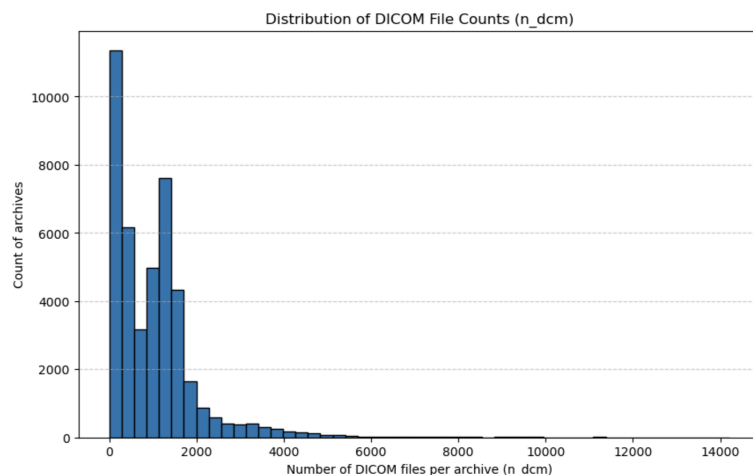


Data Preprocessing

Origin Dataset: 43161 tar files, each tar file is composed by a sequence of DICOM (Digital Imaging and Communications in Medicine) files. For a single DICOM file, it is composed by metadata and imaging data.

- Metadata refers to descriptive information of this DICOM slice, such as patient information, test information, and the parameters used in the test.
- Imaging data refers to the pixel values of the image. Most of them are stored in grey scale (1-channel), and a few of them are stored in RGB (3-channel).

Distribution of Number of DICOM Files:



Data Preprocessing:

Given the dataset is extremely large (over 42.8 million DICOM slices in total), we processed them in parallel. We divide our dataset into 309 chunks (each contains 140 tar files) and process them using 12 CPUs.

For a given tar file, we first extract the metadata and imaging data of all DICOM slices. For metadata, we take the union of all available DICOM tags across slices and flatten them into a single row dictionary for each slice. Complex tag values (lists and nested structures) are automatically converted into JSON-safe strings to ensure consistent CSV writing. Basic fields such as `tar_name`, `dcm_name`, `slice_index`, `save_path`, and `slice_status` are appended to every row.

For imaging data, we decode each DICOM slice using `pydicom` module. We convert all RGB images to grayscale to ensure the consistency of image structures, and for some DICOM slices with the data of multi-frame images, we reduced to a representative 2D

slice. Because the raw imaging data are extremely large, storing them directly in .npy format is infeasible. Instead, we first convert the pixel value into int16 format and then compress them using BLOSC module (LZ4HC + bitshuffle), which finally reduce the disk usage over 70 percent (from 25T to 8T). Given different DICOM slices in a tar file may have different height and widths, we first group these slices by image size and compress each group separately.

Finally, after preprocessing all tar files, we obtain a total of 309 chunk-level CSV files, where each CSV records the full metadata extracted from all DICOM slices within the corresponding chunk. In parallel, all imaging data from the tar files are converted into compressed int16 numpy array using BLOSC, resulting in a highly space-efficient representation of the original DICOM data.